Spin dependent Klein tunneling in altermagnets

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We take a look at a low energy altermagnetic Hamiltonian and map it to the Dirac Hamiltonian. We then use this Hamiltonian to derive spin polarized tunneling in these altermagnets.

I. THE ALTERMAGNETIC HAMILTONIAN

Altermagnets are a recently discovered phase of matter offering interesting properties that are a mix of those shown by ferromagnets and antiferromagnets. They display no net magnetization like antiferromagnets, while showing spin split bands similar to those shown by ferromagnets. We use an effective two-band, quasi planar model for our analysis, following [1]. The Hamiltonian is given by

$$H = J(\cos k_x - \cos k_y)\sigma_z + \lambda' \left[\sin\left(\frac{k_x + k_y}{2}\right)\sigma_x + \sin\left(\frac{k_y - k_x}{2}\right)\sigma_y\right], \tag{1}$$

The eigenvalues of this Hamiltonian are

$$E_\pm(k_x, k_y) = \pm\sqrt{J^2(\cos k_x - \cos k_y)^2 + \lambda'^2(1 - \cos k_x \cos k_y)} \tag{2}$$

Notice that $E_\pm(0,0) = 0$. Taylor expanding about this point, we get

$$E_\pm(k_x, k_y) = \pm\frac{\lambda'}{\sqrt{2}}\sqrt{k_x^2 + k_y^2} + O(k_x^2 + k_y^2) \tag{3}$$

Ignoring higher ordered terms, we get a linear dispersion relation. This suggests the emergence of graphene-like physics. The low energy Hamiltonian, then, is

$$H = \frac{\lambda'}{\sqrt{2}}\left[\left(\frac{\sigma_x - \sigma_y}{\sqrt{2}}\right)k_x + \left(\frac{\sigma_x + \sigma_y}{\sqrt{2}}\right)k_y\right], \tag{4}$$

Defining $S_x = \frac{\sigma_x - \sigma_y}{\sqrt{2}}$ and $S_y = \frac{\sigma_x + \sigma_y}{\sqrt{2}}$, we can write the Hamiltonian concisely as

$$H = \frac{\lambda'}{\sqrt{2}}\mathbf{k}\cdot\mathbf{S} \tag{5}$$

which has the same form as graphene's Hamiltonian. We have $\mathbf{S} = (S_x\ S_y)$ instead of $\sigma = (\sigma_x\ \sigma_y)$. This is not an issue, however, because S_x and S_y anti-commute and $S_x^2 = I$ and $S_y^2 = I$. Thus, they form a valid pair [2]. Because of the arbitrariness of the choice of σ for graphene, the solutions of the Hamiltonian remain the same up to a rotation of coordinates, since there exists a unitary matrix U such that $U^\dagger\sigma_x U = S_x$ and $U^\dagger\sigma_y U = S_y$. Under such a transformation, $(k_x\ k_y) \longrightarrow \left(\frac{k_x+k_y}{\sqrt{2}}\ \frac{k_y-k_x}{\sqrt{2}}\right)$, which corresponds to a $\frac{\pi}{4}$ rotation. Where these systems differ is the origin of their degrees of freedom. For both systems, the degrees of freedom are two. In graphene, this was because of the mismatch of the two sublattices that make up the whole lattice. For the altermagnet under consideration, it is the spin degree of freedom. This leads to some interesting physics that we will see in the next section.

II. SPIN DEPENDENT TUNNELING

We consider the problem of Klein tunneling for electrons obeying the above Hamiltonian. We place the barrier at an angle of $\frac{\pi}{4}$ with the x axis. We now do the aforementioned unitary transformation to transform our basis into one where the barrier is along the y axis. In this basis, the Hamiltonian becomes

$$H = \lambda\mathbf{k'}\cdot\sigma \tag{6}$$

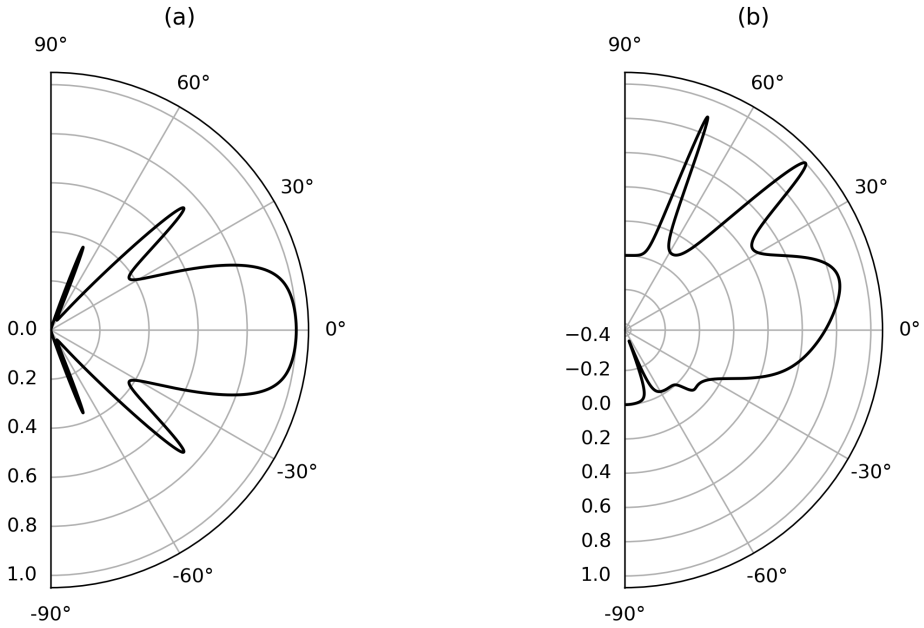


FIG. 1: (a). Net spin transmission when spin is measured along the x axis for $n = 0.5 \times 10^{16}$ m^{-2}, $p = 1 \times 10^{16}$ m^{-2} and barrier width, $D = 100$ nm. (b). Net spin transmission for spin measured at an angle of 42.75° along the x axis for the same parameters. A net spin transmission of -0.37 is observed at 68.85°. Conventionally, spin-up transmission is taken to be positive.

where **k'** is the same vector as **k**, but expressed in the new co-ordinates. We have also absorbed the $\sqrt{2}$ factor into λ' and will now use λ instead for the rest of the discussion. We define the potential

$$V(x) = \begin{cases} V_0, & 0 < x < D, \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

The Hamiltonian then becomes

$$H = \lambda \mathbf{k'} \cdot \sigma + V(x) \tag{8}$$

The eigenvalue equation is

$$H|\psi\rangle = E|\psi\rangle \tag{9}$$

This problem was the subject of [3], the results of which we will use here. We will be using the same notation as the referenced paper. So, k_F is the Fermi wavevector, $k_x = k_F \cos\phi$ and $k_y = k_F \sin\phi$ are the x and y wavevector components outside the barrier and ϕ is the angle of incidence, $q_x = \sqrt{\frac{(E-V_0)^2}{\hbar^2 v_F^2} - k_y^2}$ is the x component of the wavevector inside the barrier, $\theta = \tan^{-1}(k_y/q_x)$ is the refraction angle, $s = \text{sgn}(E)$ and $s' = \text{sgn}(E - V_0)$. The wavefunction to the right of the barrier is

$$|\psi\rangle = \psi(x,y)|\varphi\rangle = \frac{te^{ik_x x + ik_y y}}{\sqrt{2}} \begin{pmatrix} 1 \\ se^{i\phi} \end{pmatrix} \tag{10}$$

t denotes the transmission coefficient, $\psi(x,y)$ denotes the spatial part and $|\varphi\rangle$ denotes the normalized spin part of the wavefunction. Given that σ is the spin operator, the asymmetry of the two components suggests that it might be possible to have spin dependent tunneling. Let us consider the spin along the x direction, represented by the eigenvalues and eigenstates of σ_x. Denoting the eigenstates of σ_x as $|\uparrow\rangle$ (spin-up; the corresponding eigenvalue is 1) and $|\downarrow\rangle$ (spin-down; the corresponding eigenvalue is -1). The spin part is

$$|\varphi\rangle = a|\uparrow\rangle + b|\downarrow\rangle \tag{11}$$

Statistically, if N electrons were shot at the barrier, $|a|^2N$ of them would be spin-up and $|b|^2N$ would be spin-down. $|a|^2$ represents the condition probability that an electron is in the spin-up state, given that it managed to cross the barrier. The spin-up transmission is, thus, proportional to $|a|^2T$ and the spin-down transmission proportional to $|b|^2T$, where $T = |t|^2$ is the probability of transmission. It is straightforward to compute a and b. We are interested in the difference between the spin-up and spin-down transmission because it leads to the generation of spin-current.

$$(|a|^2 - |b|^2)T(\phi) = sT(\phi)\cos\phi \tag{12}$$

For positive energies, $s = 1$ and the difference is proportional to $\cos(\phi)$. We can extend this same calculation for spins along other directions. Note that the spin along the z direction gives the same conditional probability for both spin-up and down. So, measuring spin purely along the z axis does not give us anything interesting. We consider a general spin operator, the form of which is

$$\sigma_{\alpha,\beta} = \sin(\beta)\cos(\alpha)\sigma_x + \sin(\beta)\sin(\alpha)\sigma_y + \cos(\beta)\sigma_z \tag{13}$$

Performing the same calculation for this spin operator gives us a spin-current proportional to $sT(\phi)\sin(\beta)\cos(\phi - \alpha)$. There are three consequences of this result. The first is that it is possible to amplify the spin-current in a particular direction, say ϕ_0 along the x axis, up to a maximum of $T(\phi_0)$ by simply measuring spin along that direction, i.e. setting $\alpha = \phi_0$. This comes at the expense of hampering net spin transmission along other directions. Only some directions are worth amplifying, however, since there is zero transmission along certain directions. The second is that for all spin measuring directions other than the x axis, we can generate negative spin currents simply by switching the direction of the flow of electrons, as demonstrated in Fig. 1(b). The third is that it is possible to scale the transmission probabilities along all directions in $x - y$ plane by changing the angle of spin measurement along the z axis.

III. HIGHER ORDERED HAMILTONIAN

We saw a lot of interesting physics in the previous section, but a big chunk of it was lost by neglecting higher ordered terms in the Hamiltonian in section I. This time, we shall consider terms up to second order in k. This gives us

$$H = \lambda(k_x\sigma_x + k_y\sigma_y) + Jk_xk_y\sigma_z + V(x) \tag{14}$$

where we have added in the same potential from the last section. The energy and V_0 are connected by

$$E = \pm\sqrt{\lambda^2k_f^2 + J^2k_x^2k_y^2} \tag{15}$$

$$E - V_0 = \pm\sqrt{\lambda^2q_f^2 + J^2q_x^2k_y^2} \tag{16}$$

As always, we look for plane wave solutions. For our tunneling problem, the spinor components, ψ_1 and ψ_2 are

$$\psi_1 = \begin{cases} (e^{ik_xx} + re^{-ik_xx})e^{ik_yy}, & x < 0, \\ (ae^{iq_xx} + be^{-iq_xx})e^{ik_yy}, & 0 < x < D, \\ te^{ik_xx+ik_yy}, & x > D, \end{cases} \tag{17}$$

$$\psi_2 = \begin{cases} (Ae^{ik_xx+i\phi} - \frac{r}{A}e^{-ik_xx-i\phi})e^{ik_yy}, & x < 0, \\ (aBe^{iq_xx+i\theta} - \frac{b}{B}e^{-iq_xx-i\theta})e^{ik_yy}, & 0 < x < D, \\ tAe^{ik_xx+ik_yy+i\phi}, & x > D, \end{cases} \tag{18}$$

where $A = \frac{E-Jk_xk_y}{\lambda k_f}$ and $B = \frac{E-V_0-Jq_xk_y}{\lambda q_f}$. Mandating continuity at $x = 0$ and $x = D$ yields

$$r = \frac{2ie^{iq_xD}\sin(q_xD)\left[1 - A^2e^{2i\phi} + ABe^{i(\phi+\theta)} - \frac{A}{B}e^{i(\phi-\theta)}\right]}{2(1 - e^{2iq_xD}) + \frac{1}{AB}e^{-i(\phi+\theta)} + ABe^{i(\phi+\theta)} + e^{2iq_xD}\left[\frac{B}{A}e^{i(\theta-\phi)} + \frac{A}{B}e^{i(\phi-\theta)}\right]} \tag{19}$$

Unlike what's usually true, $T \neq 1 - |r|^2$ here since the probability current resulting from this Hamiltonian is different from the first order system. We compute the current in the same manner as we do in non-relativistic quantum mechanics, which is by writing down the continuity equation for probability

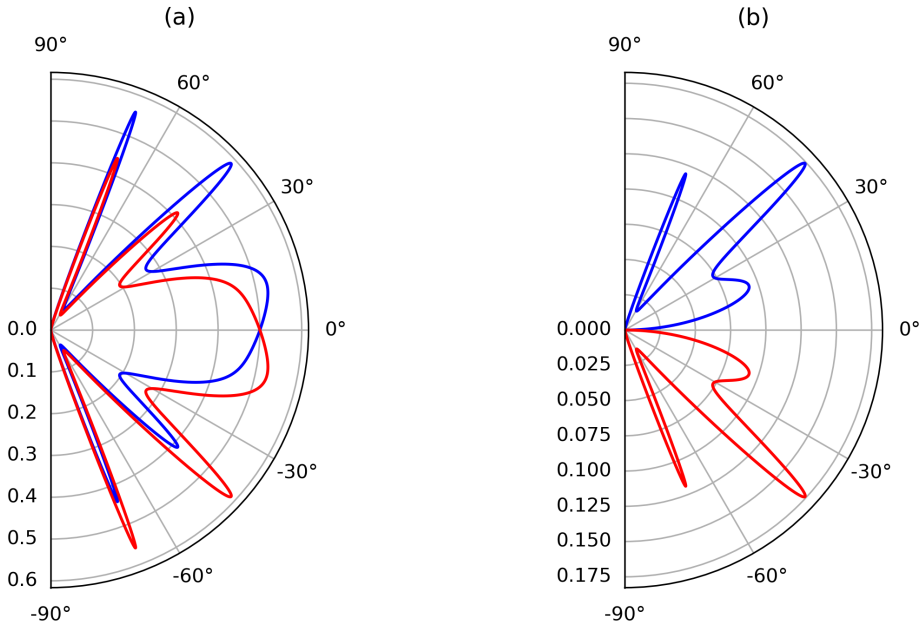


FIG. 2: (a). Spin-up transmission (blue) and spin-down transmission (red) for spins along the z axis for $\frac{J}{\lambda} = 20$. (b). Net spin transmission for spin along z axis. The (blue) red part indicates that spin (up) down dominates for the same $\frac{J}{\lambda}$ ratio. All the other parameters for both of these figures are the same as those of Fig. 1.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \mathbf{j} = 0 \tag{20}$$

where $\rho = \psi^\dagger \psi$. Using, we calculate $\mathbf{j}$ to be

$$\mathbf{j} = \frac{\lambda}{\hbar}\left[\psi^\dagger \sigma_x \psi\, \hat{\mathbf{x}} + \psi^\dagger \sigma_y \psi\, \hat{\mathbf{y}}\right] + \frac{iJ}{\hbar}\left[\frac{\partial \psi^\dagger}{\partial y}\sigma_z \psi\, \hat{\mathbf{x}} - \psi^\dagger \sigma_z \frac{\partial \psi}{\partial x}\, \hat{\mathbf{y}}\right] \tag{21}$$

We now calculate the incident (j_i), reflected (j_r) and transmitted (j_t) currents along the x direction and demand that $|j_i| = |j_r| + |j_t|$. This produces

$$j_i = \frac{2\lambda R_1 \cos\phi + Jk_y(1 - R_1^2)}{\hbar} \tag{22}$$

$$j_r = |r|^2 \frac{Jk_y(1 - R_2^2) - 2\lambda R_2 \cos\phi}{\hbar} \tag{23}$$

$$j_t = |t|^2 \frac{2\lambda R_1 \cos\phi + Jk_y(1 - R_1^2)}{\hbar} \tag{24}$$

which finally gives us

$$T = 1 - \left|\frac{j_r}{j_i}\right| \tag{25}$$

This finally allows us to compute T. A major consequence of this new Hamiltonian is the possibility of spin current generation when spin is measured along the z axis. Repeating the same calculation as the previous section, but with the eigenvectors of σ_z this time $T_\uparrow = T\frac{1}{1+A^2}$, $T_\downarrow = T\frac{A^2}{1+A^2}$ and $\Delta T = T_\uparrow - T_\downarrow = T\frac{1-A^2}{1+A^2}$. In Fig. 2, we plot these transmissions. Unlike with x axis spins, we can't have a spin current along the x axis for z spins since $A = 1$ fo $\phi = 0$. Clearly, the direction of the travelling wave decides which spin dominates. Given that, the symmetry between the upper half and the lower planes in Fig. 2(b) is fairly intuitive.

IV. CONSTANT ENERGY CONTOURS

In the previous section, we implictly made the approximation that $|k|$ is constant. Looking back at (15), it is clear that this holds exactly when $J = 0$ and approximately when $J << \lambda$. In this section, we consider the exact case. Fig. 2 shows the constant energy contours implied by (15), along with the expectation value of the spin along z axis, $S_z = \frac{Jk_xk_y}{E}$. Clearly, a higher J shows more pronounced spin splitting.

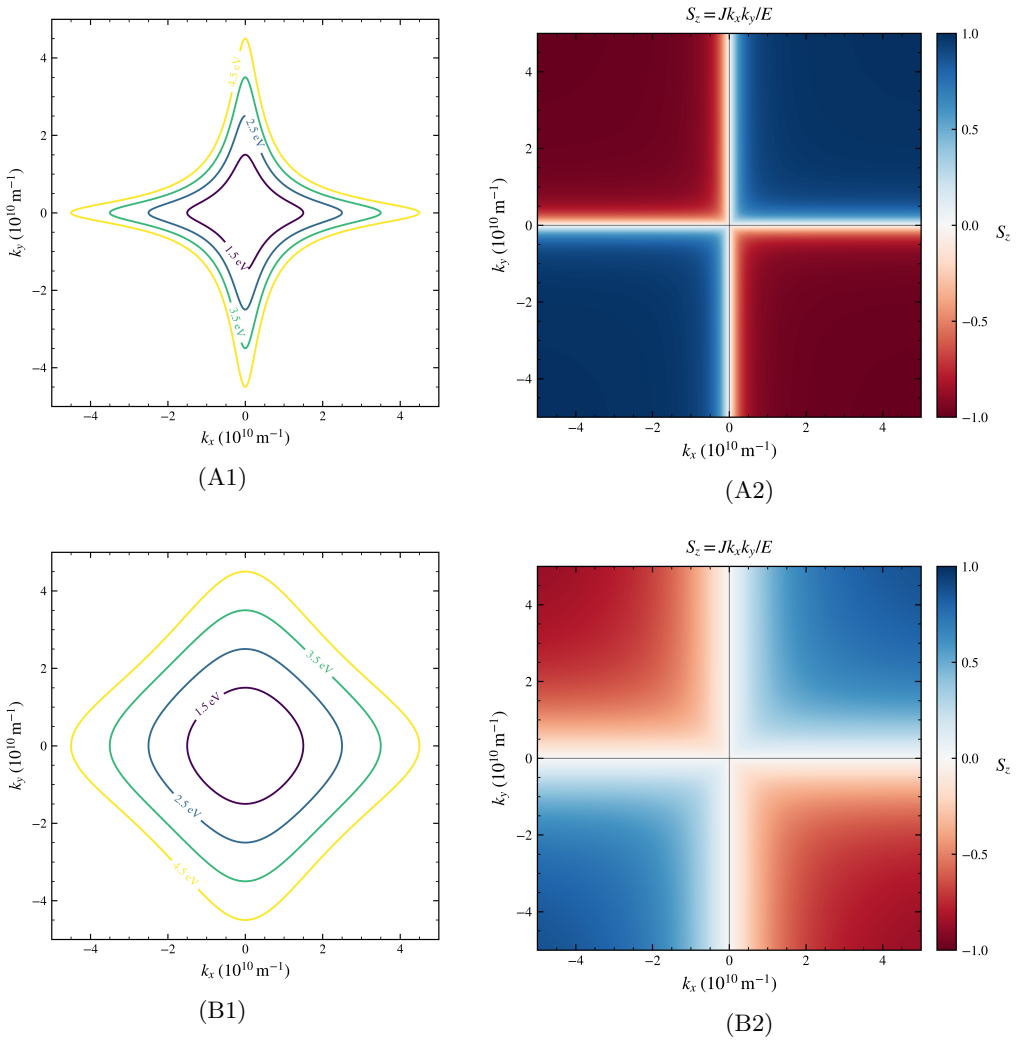


FIG. 3: $\lambda = 1$ eV for all 4 figures, whereas $J = 3$ eV for (A1) and (A2) and $J = 0.5$ eV for (B1) and (B2). (A1) and (B1) depict constant energy contours. These energies are (from the innermost to the outermost contour) 1.5 eV, 2.5 eV, 3.5 eV and 4.5 eV. (A2) and (B2) show the expectation value of the spin along the z-axis, ignoring the $\frac{\hbar}{2}$ factor.

Let us now consider the tunneling problem again. This time, we shall fix E and V_0. This has an interesting consequence. Recall that, earlier, we had fixed n and p, which determined k_f and q_f. These, along with the incidence angle, determined both E and V_0. However, once we fix E and V_0, some incidence angle may be impossible to observe. From (15) and (16), it is clear that k_y is restricted in what values it can take.

$$k_{y_{max}} = \frac{\min(|E|, |E - V_0|)}{\lambda} \tag{26}$$

This is to ensure that $k_x^2, q_x^2 \geq 0$. If $|E - V_0| < |E|$, then $k_{y_{max}} < \frac{|E|}{\lambda}$, which means that $|k_x|$ is bounded below by $k_{x_{min}}$.

$$k_{x_{min}} = \sqrt{\frac{E^2 - \lambda^2 k_{y_{max}}^2}{\lambda^2 + J^2 k_{y_{min}}^2}} \tag{27}$$

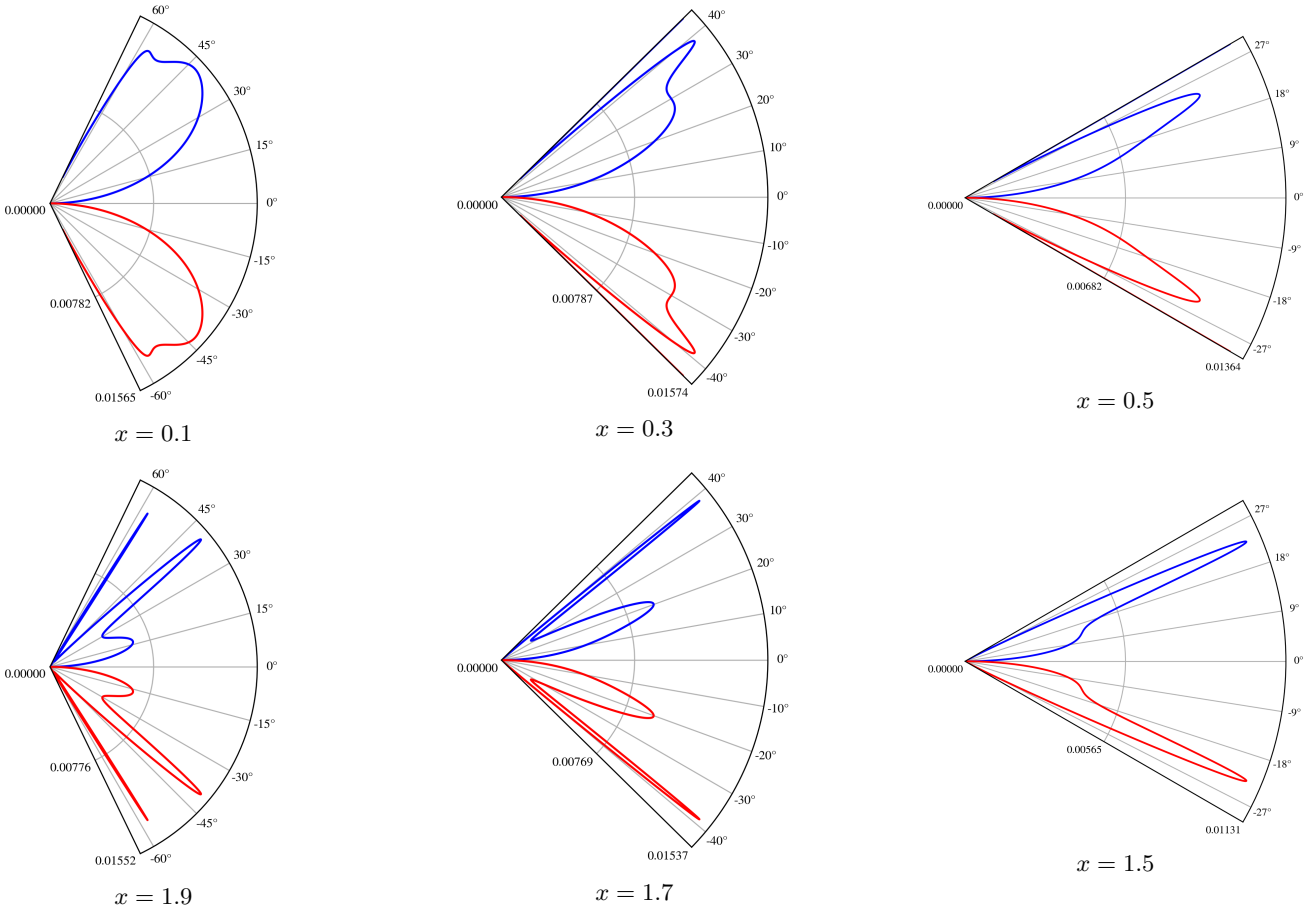


FIG. 4: Depictions of net spin transmission for $E = 10$ meV and $V_0 = xE$ for $\frac{J}{\lambda} = 3$.

The maximum allowed incidence angle is then

$$\phi_{max} = \tan^{-1}(k_{y_{max}}/k_{x_{min}}) \tag{28}$$

The various plots in Fig. 3 all satisfy $|E - V_0| < |E|$ and it illustrates the angular range limitation as a result. When $E = V_0$, the angular range becomes 0 and the problem becomes completely one dimensional, since $k_y = 0$.

V. REFERENCES

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